

Weather & Air Quality Station

Chain K — Hardware & Systems Foundations · Full written lesson

This is the project that actually lives outside — everything else in Chain K can sit on a desk. Weather stations force real decisions about power, waterproofing, and long-running reliability, not just circuit theory.

Lesson 1 — Wiring the BME280 and Rain Gauge, and Trusting the Readings

Getting a believable number out of a sensor is a different skill from wiring it up — and a weather station is the project in this chain where "believable" actually matters, not just "changes."

- The BME280 is one cheap I2C breakout covering temperature, humidity, and pressure — but clones exist that drift badly over time compared to a genuine, well-documented part.
- A tipping-bucket rain gauge produces one pulse per tip via a reed switch. The raw pulse count means nothing on its own — it only converts to real rainfall depth through a stated calibration figure (commonly 0.2-0.5mm per tip).
- Optional air-quality add-ons — a PMS5003 (PM2.5 particulate) or SCD40 (CO₂) — bolt onto the same I2C/UART pattern once the core sensors are working.

The trap: a rain gauge sold with no calibration figure. Without it, the sensor can only ever tell you "some rain happened," never how much.

Worked example: a rain gauge with a stated 0.2mm/tip calibration converts a raw tip-count directly into a real depth of rainfall (10 tips = 2.0mm); an uncalibrated gauge with the identical mechanism can only report "10 tips happened."

Lesson 2 — Sizing Solar Power for the Cloudiest Week, Not the Sunniest Day

This is the actual engineering decision this project is built around: a panel that looks comfortably oversized on a sunny day can still leave a station dead after a real cloudy week.

- Daily consumption: a station's average current draw times 24 hours gives a real daily mAh budget.
- Daily harvest: panel watts times real sun-hours (not daylight hours) times a real-world efficiency derate, converted to mAh at the battery's voltage.
- The actual design target is the net daily balance staying positive through the cloudiest realistic week — not just an average or sunny day.
- If running a real deficit, autonomy — usable battery capacity divided by the daily deficit — gives the real number of days until the low-voltage cutoff kicks in.

The trap: battery undersized for winter's shorter solar days. The same setup that nets a healthy surplus on a summer day can run a real deficit on a winter day.

Worked example: a 1W panel and a 30mA average draw nets roughly +496mAh/day on a sunny (6 sun-hour) day, but the identical setup nets roughly -517mAh/day on a cloudy (1 sun-hour) day — leaving only about 3 days of autonomy on a 2000mAh battery.

Lesson 3 — Choosing Wifi vs. LoRa for a Remote, Low-Bandwidth Node

This node has an unusually easy connectivity decision once you look at what it actually needs to send: a handful of numbers, occasionally — not a live video stream.

- Wifi offers higher bandwidth and dead-simple integration, but real range is limited and it's power-hungry relative to LoRa.
- LoRa trades bandwidth for much longer range and far lower power — a great match for infrequent, tiny payloads.
- The real decision isn't "which radio is generally better" — it's "does wifi actually reach this exact mounting spot."

The trap: wifi range from an outdoor mounting spot back to the router. LoRa avoids this entirely if wifi doesn't reach.

Worked example: the same station moved from a spot with full wifi bars to the far end of the yard can lose the connection entirely; swapping to LoRa fixes exactly that problem, with no other design change needed.

Lesson 4 — Weatherproofing Correctly, and Building for Months Unattended

This project reuses Walkie-Talkie-Build's exact waterproofing pattern, but adds one detail specific to a barometric sensor that trips people up.

- The BME280 needs a vent membrane, not a sealed wall. A fully sealed enclosure traps a fixed volume of air, and its pressure drifts from the real outside pressure as temperature changes.
- The same vent membrane that lets pressure equalize also reduces condensation.
- Long-term unattended reliability means power, connectivity, and weatherproofing all have to keep working together continuously, with nobody checking on it for weeks.

The trap: a fully sealed enclosure reading barometric pressure wrong.

Worked example: the BME280 sits behind a vent membrane rather than a sealed wall, specifically because a fully sealed box reads pressure wrong, not only because of condensation risk.

Reflection

If your station's battery only has 3 days of autonomy through a cloudy week, what are your actual options, and how would you decide between them? Why does an outdoor, unattended project force real engineering decisions that a desk-bound project can defer indefinitely?